

CMAN-9

TASKS:

1. Dominant kernel

Kernel: Scaled Dot-Product Self-Attention

Operations: QK^T matmul + AV weighted sum

Dimensions: $N=512$, $d-k=64$

Data Types: INT8 inputs, 32-bit accumulator,
32-bit output all are signed.

INT8 - 1 Byte per element

32-bit - 4 Bytes

Q-Matrix: $N \times d-k = 512 \times 64$

K-Matrix: $N \times d-k = 512 \times 64$

V-Matrix: $N \times d-k = 512 \times 64$

Attention Score Matrix: $N \times N = 512 \times 512$

2. FLOPS Count

Each multiply-accumulate (MAC) = 2 FLOPS (1 multiply + 1 Add)

QK^T Matrix Multiply:

For each of N output rows, compute dot-product of length $d-k$.

$$\begin{aligned}
 \text{FLOPS}_{\text{QKT}} &= 2 \times N \times N \times d-k \\
 &= 2 \times 512 \times 512 \times 64 \\
 &= 33,554,432 \text{ FLOPS}
 \end{aligned}$$

AV weighted sum:

For each of N output rows, compute weighted sum over N attention weights applied to $d-k$ dimensional V vectors

$$\begin{aligned}
 \text{FLOPS}_{\text{AV}} &= 2 \times N \times N \times d-k \\
 &= 2 \times 512 \times 512 \times 64 \\
 &= 33,554,432 \text{ FLOPS}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total FLOPS} &= \text{FLOPS}_{\text{QKT}} + \text{FLOPS}_{\text{AV}} \\
 &= 33,554,432 + 33,554,432 \\
 &= 67,108,864 \text{ FLOPS} \\
 &= 67.1 \text{ MFLOPS} \\
 &=
 \end{aligned}$$

3. Bytes Transferred

Case A: No data reuse (lower bound)

All matrices loaded fresh from off-chip memory every invocation.

$$\text{INT8} = 1 \text{ Bytes per element}$$

$$\begin{aligned}
 \text{Bytes-}Q &= N \times d-k \times 1 = 512 \times 64 \times 1 = 32,768 \text{ Bytes} \\
 \text{Bytes-}k &= N \times d-k \times 1 = 512 \times 64 \times 1 = 32,768 \text{ Bytes} \\
 \text{Bytes-}V &= N \times d-k \times 1 = 512 \times 64 \times 1 = 32,768 \text{ Bytes} \\
 \text{Bytes-out} &= N \times d-k \times 4 = 512 \times 64 \times 4 = 131,072 \text{ Bytes} \\
 &\quad (32\text{-bit output})
 \end{aligned}$$

$$\begin{aligned}
 \text{Bytes-no-reuse} &= \text{Bytes-}Q + \text{Bytes-}k + \text{Bytes-}V + \text{Bytes-out} \\
 &= 32,768 + 32,768 + 32,768 + 131,072 \\
 &= 229,376 \text{ Bytes} \\
 &= \underline{\underline{224 \text{ KB}}}
 \end{aligned}$$

Base B: Perfect on-chip Data Reuse (upper Bound)

k and V are loaded once and cached in on-chip SRAM.

Only Q is streamed from off-chip memory per invocation.

Outputs are written back to off-chip memory

$$\begin{aligned}
 \text{Bytes-}Q &= N \times d-k \times 1 = 32,768 \text{ Bytes} \\
 \text{Bytes-Out} &= N \times d-k \times 4 = 131,072 \text{ Bytes} \\
 k, V &= 0 \text{ Bytes}
 \end{aligned}$$

$$\begin{aligned}
 \text{Bytes-full-reuse} &= \text{Bytes-}Q + \text{Bytes-out} \\
 &= 32,768 + 131,072 \\
 &= \underline{\underline{160 \text{ KB}}}
 \end{aligned}$$

4. Arithmetic Intensity:

$$\begin{aligned}\text{AI-lower} &= \text{FLOPs-dominant} / \text{Bytes-no-reuse} \\ &= 67,108,864 / 229,376 \\ &= 292.6 \text{ FLOP/Byte}\end{aligned}$$

$$\begin{aligned}\text{AI-Upper} &= \text{FLOPs-dominant} / \text{Bytes-full-reuse} \\ &= 67,108,864 / 163,840 \\ &= 409.6 \text{ FLOP/Byte}\end{aligned}$$

Sky 130A ridge point = $64 \text{ GFLOPs} \div 8 \text{ GB/s} = 8.0 \text{ FLOP/Byte}$
Both bounds are Compute Bound.

5. Bottleneck and Improvement

Current Bottleneck: Hardware Interface Bandwidth
AXI4-stream at 250 MHz delivers 6 GB/s effective

Ridge point = 8.0 FLOP/Byte

at AI = 42.7 design is interface-bound

Highest Leverage Improvement: Widen AXI4-stream bus
from 256-bit to 512-bit

- double bandwidth from 6 to 12 GB/s
- Pushes ridge point to 5.3 FLOP/Byte
- design becomes fully compute-bound
- achieves full 64 GFLOPs/Reale.